

OPERATING PREMISE AND DAYS 1-30

Earn the right to standardize.

The first objective is not to introduce a new architecture regime. It is to understand where Thermo Fisher's existing standards already accelerate scientific and business teams, where legitimate variation must remain, and where recurring friction signals a missing enterprise capability.

What should be materially different by day 120

|                                                                                                                                                                      |                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>One shared architecture map</b><br>Platforms, decision rights, active demand, risk boundaries, review friction, and ownership are legible enough to guide action. | <b>One bounded proof</b><br>A decision-critical AI use case has explicit outcome, authority, control, integration, operating, adoption, and evidence contracts.          |
| <b>One reusable pattern</b><br>The proof produces a documented disposition and assets another squad can adopt, adapt, or knowingly decline.                          | <b>One operating cadence</b><br>Architecture decisions, exceptions, cost, reliability, adoption, and learning are reviewed through a practical PPI-compatible mechanism. |

Days 1-30 - Read the system before prescribing

|                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                       |
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| <b>Stakeholder topology</b><br>Business and scientific sponsors; enterprise, solution, data, cloud, integration, and security architects; privacy, compliance, legal, quality, and risk leaders; product, automation, squad, operations, support, and FinOps leaders. | <b>Evidence baseline</b><br>Inventory active AI demand. Trace one request from intake through operation. Baseline review latency, recurring exceptions, duplicated components, reliability events, cost visibility, retained use, and unresolved ownership. Separate facts from discovery hypotheses. |
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Day-30 deliverable register

| Deliverable                              | Decision it enables                                                      | Validation partner                               |
|------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------|
| Architecture landscape and ownership map | Where standards, platforms, and decision rights are clear or conflicting | Architecture and platform leaders                |
| AI demand and friction portfolio         | Where enterprise attention creates the most reusable value               | Business, product, squad, and governance leaders |
| Evidence baseline                        | What must improve and how improvement will be observed                   | Operations, support, FinOps, quality, and risk   |
| Candidate proof shortlist                | Which bounded use cases merit deeper qualification                       | Accountable business and technical owners        |

Day-30 exit criteria and questions

|                                                                                                                                                                                                                                      |                                                                                                                                                                                                                       |
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| <b>Exit criteria</b><br>The landscape and demand portfolio are credible enough to act; ownership conflicts are visible; the proof shortlist is evidence-based; and the current paved roads and legitimate exceptions are understood. | <b>Questions still open</b><br>Where do teams route around standards? Which controls arrive too late? Which platforms already earn trust? What recurring friction should become an architecture-product backlog item? |
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DAYS 31-90

# Choose the proof. Then run the pattern under real load.

The first proof should expose real architecture tradeoffs, be bounded enough to finish responsibly, and be reusable enough to leave more than a one-off success behind. Technical design, governance, operations, economics, and adoption must be exercised together.

## Proof-selection criteria

| Lens                        | Qualification question                                                                  | Disqualifying signal                                         |
|-----------------------------|-----------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Outcome and consequence     | Named workflow, decision, owner, baseline, and observable result?                       | Enthusiasm without accountable outcome ownership             |
| Data and authority          | Sources, entitlements, human decision rights, abstention, and escalation identifiable?  | Unclear data authority or autonomous action without an owner |
| Enterprise reuse            | Likely to recur across products, functions, or scientific workflows?                    | A bespoke edge case with little transferable learning        |
| Integration and operability | Interfaces, observability, incidents, rollback, cost, and lifecycle ownership explicit? | No credible production or support owner                      |
| Adoption burden             | Can affected people test, challenge, learn, and retain the changed workflow?            | Success depends on communications rather than changed work   |

## Six boundaries in the architecture contract

|                                                                                                                  |                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>1. Outcome</b><br>Workflow, decision, consequence, owner, baseline, and measurement.                          | <b>2. Authority</b><br>Authoritative context, inference, human judgment, permitted actions, abstention, and escalation. |
| <b>3. Control</b><br>Privacy, security, compliance, quality, lineage, retention, evaluation, and audit evidence. | <b>4. Integration</b><br>APIs, events, microservices, data products, ERP, CRM, instruments, RPA, and dependencies.      |
| <b>5. Operations</b><br>SLOs, telemetry, drift, incidents, rollback, capacity, unit cost, and support ownership. | <b>6. Reuse</b><br>Pattern owner, versioning, documentation, exceptions, adoption support, and promotion evidence.      |

## Four synchronized delivery workstreams

|                                                                                                                                                                                             |                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Solution and integration</b><br>Components, interfaces, data flows, prompts, retrieval, model routing, orchestration, failure paths, reversible choices, and fit with existing services. | <b>Trust and control</b><br>Workflow-based evaluation, access, lineage, privacy, security, retention, audit evidence, human authority, abstention, escalation, and proportional red-team tests. |
| <b>Run and economics</b><br>Traces, logs, metrics, latency, cost, capacity, incidents, recovery, SLOs, rollback, release controls, support demand, and lifecycle accountability.            | <b>Adoption and workflow change</b><br>User testing, exception behavior, trust gaps, training, workarounds, retained use, standard work, support paths, and feedback-to-change.                 |

## Review cadence and day-90 evidence

| Cadence                  | Review focus                                                           | Decision / artifact                                                                                 |
|--------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Twice weekly             | Interfaces, blockers, tests, controls, and failure modes               | Resolve, escalate, or revise                                                                        |
| Weekly                   | Outcome, authority, trust, operability, economics, adoption, and reuse | Proceed, narrow, hold, or stop                                                                      |
| Biweekly sponsor readout | Business consequence, risk posture, decisions required, and learning   | Protect focus and remove constraints                                                                |
| End of phase             | Evidence against contract and pre-mortem                               | Architecture, evaluation, operating, and adoption records; promote, revise, hold, retire, or exempt |

DAYS 91-120

# Turn the proof into an enterprise learning system.

The final phase is not a rollout announcement. It is the conversion of evidence into reusable capability, explicit decision rights, a governed exception path, and a visible backlog of architecture-product improvements.

## Assets that should survive the proof

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| <b>Reference architecture</b><br>Approved components, seams, interfaces, data authority, control points, deployment options, and legitimate variation. | <b>Delivery kit</b><br>Templates, IaC guidance, evaluation harnesses, control requirements, integration contracts, and example decision records. |
| <b>Run kit</b><br>Telemetry, SLOs, alerts, incident and rollback playbooks, cost model, capacity guidance, ownership, and retirement criteria.         | <b>Adoption kit</b><br>Workflow changes, role guidance, training, review procedures, support path, feedback loop, and retained-use measures.     |

## Decision-rights model to validate

| Actor                                            | Primary decision right                                                            | Evidence owed to others                                                   |
|--------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Enterprise architecture                          | Principles, reusable patterns, exceptions, lifecycle, and coherence               | Rationale, usable guidance, and visible tradeoffs                         |
| Platform and data owners                         | Paved-road capabilities, service boundaries, reliability, security, and economics | SLOs, constraints, cost, roadmap, and support commitments                 |
| Product and delivery squads                      | Workflow design, implementation inside the contract, and local evidence           | Outcome, tests, integration behavior, incidents, adoption, and exceptions |
| Privacy, security, quality, compliance, and risk | Control sufficiency proportional to consequence                                   | Specific requirements, decision timing, and exception rationale           |
| Business or scientific owner                     | Outcome priority, human authority, adoption, and residual-risk acceptance         | Named owner, baseline, consequence, and retained-use evidence             |

## Balanced architecture scorecard

| Dimension             | Initial evidence - not an invented target                          | Leadership question                                    |
|-----------------------|--------------------------------------------------------------------|--------------------------------------------------------|
| Outcome               | Cycle time, quality, throughput, customer or scientific result     | Did the workflow improve enough to justify the burden? |
| Reuse                 | Adoption of components or patterns; duplication avoided            | What can another team safely reuse?                    |
| Reliability and trust | SLOs, incidents, recovery, grounding, access, controls, escalation | Does evidence match the consequence?                   |
| Economics             | Unit cost, infrastructure, support effort, and value category      | Is the capability economically sustainable?            |
| Adoption and learning | Retained use, exception behavior, feedback-to-change time          | Did the system change work and improve through use?    |

## Day-120 leadership readout

- Disposition of the bounded proof: promote, revise, hold, retire, or exempt - with evidence and unresolved risks.
- Reference assets ready for use, owners assigned, and the exception path documented.
- Architecture-product backlog ranked by recurring squad friction, enterprise risk, and reuse potential.
- Next-quarter priorities tied to outcome, reliability, trust, economics, adoption, and learning evidence.

**Discovery boundary:** specific internal platforms, standards, organization design, baselines, and target values must be validated with Thermo Fisher leaders. This plan defines the questions, mechanisms, and evidence - not presumed knowledge of undisclosed internal conditions.